

1 Intro to PDE

1.1 Basic Concepts

Notation: Using $x^1, \dots, x^n$ to denote $x_1, \dots, x_n$. Sometimes using x, y, z, t to denote variables. $\frac{\partial}{\partial x^i} u = u_{x^i} = \partial_i u = \partial_{x^i} u$

PDE: Order: highest order of derivative. **Linearity:** linear in u and its derivatives. **variable|constant coeff:** 是否函数系数有变量.

Linear differential operator ($\mathcal{L}$): If $\mathcal{L}(\alpha u + \beta v) = \alpha \mathcal{L}(u) + \beta \mathcal{L}(v)$. ps: $\mathcal{L}$ 不含函数本身 (e.g. $\mathcal{L} = \partial_t - \Delta$); $\mathcal{L}$ 作用的 u, v 是函数, $\alpha, \beta \in \mathbb{R}$ 是常数.

Superposition Principle: If $u_1, \dots, u_m$ are sols of linear homogeneous PDE: $\mathcal{L}(u) = 0$, then $u = \sum_{i=1}^m c_i u_i$ is also a sol.

Relating non-homogeneous PDE: If S_H sols of $\mathcal{L}(u) = 0$, u_p particular sol of $\mathcal{L}(u) = f$, then sols of $\mathcal{L}(u) = f$ is $S = \{u_p + u_h : u_h \in S_H\}$.

Direction Derivative: For $u : \mathbb{R}^n \rightarrow \mathbb{R}$, $\nabla u = (u_{x^1}, \dots, u_{x^n})$. Directional derivative of u in direction v (usually unit vector) is $\nabla u \cdot v$.

Const.|variable Coeff. Transport Equation: $^1 a \partial_x u(x, y) + b \partial_y u(x, y) = 0 \quad | \quad ^2 a(x, y) \partial_x u(x, y) + b(x, y) \partial_y u(x, y) = 0$

$^1 \Leftrightarrow \nabla u \cdot (a, b) = 0 \quad | \quad ^2 \Leftrightarrow \nabla u \cdot (a(x, y), b(x, y)) = 0$ (i.e. u 在随时斜率为 (a, b) 的直/曲线上为常数 $\Rightarrow$ 减少一个变量, u 可写为"由直/曲线"决定的函数)

$\Rightarrow ^1 u(x, y) = f(ax - by) = f(c), ax - by = c \quad | \quad ^2 \frac{dy}{dx} = \frac{b(x, y)}{a(x, y)} \Rightarrow$ 解出曲线方程得类似 $g(x, y) = c$ (一定要 constant c) $\Rightarrow u(x, y) = f(g(x, y))$

1.2 Norms and Divergence Theorem

C^k norms: Let $f : \Omega \rightarrow \mathbb{R}$ (Ω is domain: open+connected). Then $\|f\|_{C^k(\Omega)} := \sum_{\alpha=0}^k \sup_{x \in \Omega} |f^{(\alpha)}(x)|$ (ps: $0-k$ 阶级导函数, 每一个在 Ω 上 $\sup$ 的绝对值之和).

· If $\Omega \subseteq \mathbb{R}^n$, $\|f\|_{C^k(\Omega)} := 0-k$ 阶偏导函数在 Ω 上 $\sup$ 绝对值之和. (e.g. $\|f\|_{C^2(\Omega)} := \sup_{(x,y) \in \Omega} |f| + \sup_{(x,y) \in \Omega} |f_x| + \sup_{(x,y) \in \Omega} |f_y| + \sup_{(x,y) \in \Omega} |f_{xy}| + \sup_{(x,y) \in \Omega} |f_{yx}| + \sup_{(x,y) \in \Omega} |f_{xx}| + \sup_{(x,y) \in \Omega} |f_{yy}|$)

· For $f = f(x, y)$ Using $\|f\|_{C^{1,2}} := \sum_{\alpha=0}^1 \sup_{(x,y) \in \Omega} |\partial_x^\alpha f| + \sum_{\beta=1}^2 \sup_{(x,y) \in \Omega} |\partial_y^\beta f|$

L^p norms: Let $f : \Omega \rightarrow \mathbb{R}$ (Ω is domain, $\Omega \subseteq \mathbb{R}^n$). Then $\|f\|_{L^p(\Omega)} := \left(\int_\Omega |f(x)|^p d^n x \right)^{1/p}$ for $1 \leq p < \infty$ ps: $d^n x = dx_1 \cdots dx_n$

Norms Properties: ¹**Non-negative:** $\|f\|_{L^p(\Omega)} \geq 0$; ($\|f\|_{L^p(\Omega)} = 0 \Leftrightarrow f = 0$ almost everywhere.)

²**Scaling:** $\|\lambda f\|_{L^p(\Omega)} = |\lambda| \|f\|_{L^p(\Omega)}$ ³**Triangle inequality:** $\|f + g\|_{L^p(\Omega)} \leq \|f\|_{L^p(\Omega)} + \|g\|_{L^p(\Omega)}$

Divergence Theorem: Let $\mathbf{F} : \Omega \rightarrow \mathbb{R}^n$ ($\Omega \subseteq \mathbb{R}^n$ is domain). Then $\int_\Omega \nabla \cdot \mathbf{F} dx dy dz = \int_{\partial\Omega} \mathbf{F} \cdot \hat{\mathbf{N}} d\sigma$ (ps: $\hat{\mathbf{N}}$ 单位外法向量 on surface $\partial\Omega$; $\text{div}(\mathbf{F}) = \nabla \cdot \mathbf{F}$)

· **Surface integral:** If $\partial\Omega = \{(x, y, z) : z = \phi(x, y)\}$: $\int_{\partial\Omega} \mathbf{F} \cdot \hat{\mathbf{N}} d\sigma = \int_D \mathbf{F} \cdot \hat{\mathbf{N}} \sqrt{1 + |\nabla \phi|^2} dx dy = \int_D \mathbf{F} \cdot (-\phi_x, -\phi_y, 1) dx dy$ (ps: D 是 $\partial\Omega$ 在 xy 平面的投影)

2 Heat Equation

3 Laplace's and Poisson's equations

4 Wave Equation